

Project Requirements

The PROG 36944 – Advanced .NET Server-Side Technologies project is a group project where 3 students collaborate equally to create an enterprise web-based. A proposal must be submitted and accepted. Projects must incorporate the principles, technologies and techniques covered in class as well as something new which will need a certain level of research and experimentation.

Project Requirements

Content Requirements

1. A presentation tier consisting of at least 6-8 views (two views at least per team member) that are significant from the point of view of interactivity with the user. This shall allow each team member to practice the interactivity mechanisms studied in class.
2. A business logic tier consisting of at least 3-4 significant business use-cases (one significant business use-case per team member) implemented by 6-8 business logic classes (minimum two classes per team member) other than the user interface classes developed for Part 1.
3. A data tier that uses a database to persist the data.

Technology Requirements

1. Code developed using Visual Studio 2022 or VS Code
2. Design developed using Visual Paradigm or any other online modeling tools
3. Frameworks to be used
 - a. ASP.NET Core with client-side frameworks (e.g. Bootstrap, JQuery)
 - b. Web API for web services
 - c. Entity Framework Core for model persistence to database
 - d. NUnit testing framework for unit testing
4. Data formats:
 - a. SQL Server for relational database
 - b. JSON for data delivered through Web API

Principles and Best Practices

- c. Object-Oriented Design. The application shall demonstrate the application of best principles of object-oriented design:
 - i. Information hiding / encapsulation
 - ii. Code reuse through inheritance
 - iii. Reuse through abstraction and polymorphism
 - iv. Loose coupling, separation of concerns

- v. Simple, cohesive classes and methods
- d. UML Model. The design of the project shall be created in Visual Paradigm and shall contain the following types of diagrams
 - i. Use-Case diagrams to show the context diagram of the system and the business logic implemented
 - ii. UML class diagrams to show structure
 - iii. Sequence diagrams and/or activity diagrams to show interactivity between the user and the presentation tier components as well as between the presentation tier components and the business logic and data tier components
 - iv. Any other UML diagram necessary for the design and communication of the design of the system
- di. Coding conventions
 - i. MVC and coding conventions
 - ii. Scope prefixes conventions (e.g. use of `_` prefix for any field)
 - iii. Naming conventions: names start with capital letters for classes and methods and properties and start with a small letter for variables and fields.
 - iv. Descriptive names for classes, methods and variables
- dii. Commenting. The code shall be commented according to the following requirements
 - i. Each class shall have a header with the principal author of the code and a short description of what the code in the file is for
 - ii. Each method and field shall be commented with *XML style comments* and include a brief description of the purpose of method and field. If the Author of the method is different than the author of the class the author name shall also be included in method headers

Project Plan Requirements

- a. *Each C# class shall have only one author out of the group. "We all worked on it together"* is not acceptable as that would never happen in a professional team project. As a team you collaborate but each team member must take ownership and responsibility of a piece of the code. You will be graded individually based on your contribution.
- b. *The author shall be clearly documented in each source file.* Again, multiple authors in a single file not acceptable.
- c. *Work assignments* shall be clearly documented through an up-to-date *project plan* that is to be delivered with the project.

Project Presentation (Video Submission – Combined as one)

- a. Present the purpose of the application
- b. Present the application's project plan
- c. Demo the application functionality. Each student in the group shall demo the functionality he/she worked on
- d. Identify how the project completed meets the requirements identified in this document:
 - iv. Content requirements
 - v. Technology requirements
 - vi. Principles and Best Practices

Project Deliverables

1. Visual Studio Project with complete source code and any assets necessary to execute the project such as file, databases, images etc.
2. Up to date UML model created. The UML model can be integrated in the Visual Studio solution / project or delivered separately.
3. Presentation materials (e.g. PowerPoint, Sway, Prezi, etc.)
4. Final project report consisting of the updated proposal document with what was actually completed and with the differences between the initial proposal and the actual outcome highlighted. The Project Plan section should clearly identify the detailed responsibilities that each team member had in the completion of the project

All the project deliverables shall be archived and submitted via SLATE

Project Schedule

The development plan is organized in 3 milestones. The completion of each milestone will be verified and will count in the final grade of the project.

1. **Week 8:** Project Start
2. **Week 9:** Project Proposal Submitted (Inception Phase)
3. **Week 10:** Iteration 1 (Elaboration Phase, 30% completion)
4. **Week 12:** Iteration 2 (Elaboration Phase, 70% completion)
5. **Week 12:** Iteration 3 (Development Phase, 100% completion)
6. **Week 13:** Project Presentation
Delivery. Before project presentations the project completion document must be handed in.